№T1.

$\xi \sim R(0, \theta)$, $\theta > 0$

вер. модель

$\theta \in \Theta = (0; +\infty)$

Выборка $\vec{X_n}$

$\tilde{\theta}_1 = 2\vec{X} = 2 \cdot \frac{1}{n} \sum_{i=1}^{n} X_i$

$\tilde{\theta}_2 = X_{min}$

$\tilde{\theta}_3 = X_{max}$

$\tilde{\theta}_4 = X_1 + \frac{1}{(n-1)} \sum_{i=2}^{n} X_i$

$M\xi = \int_{-\infty}^{a} x \, dF(x, \theta) = \int_{0}^{\theta} x \frac{1}{\theta} dx = \frac{\theta}{2}$

$P(x, \theta) = \frac{1}{\theta} \{(0, \theta)\}$

$\xi_\theta \sim R(a, b)$ $\quad p(x) = \frac{1}{b-a} \{(a, b)\}$

$M\xi^2 = \int_{0}^{\theta} x^2 \frac{1}{\theta} dx = \frac{\theta^2}{3}$

$D\xi = M\xi^2 - (M\xi)^2 = \frac{\theta^2}{3} - \frac{\theta^2}{4} = \frac{\theta^2}{12}$.

Парам-я модель.

Несм. $\forall \theta \in \Theta \quad M\tilde{\theta} = \theta$

ост. $\forall \theta \in \Theta \quad \forall \varepsilon > 0 \quad P(|\tilde{\theta} - \theta| \geq \varepsilon) \xrightarrow[n \to \infty]{} 0$.

1) $\tilde{\theta}_1$

$\forall \theta > 0 \quad M\tilde{\theta}_1 = M\left(2 \frac{1}{n} \sum X_i\right) \ominus$

X_i назыв. случ. велич. $X_i \sim R(0,\theta)$

① $\frac{2}{n}\sum MX_i = \frac{2}{n} nM\xi = \theta$ несм.

$$D\tilde{\theta}_1 = D\left(2\frac{1}{n}\sum X_i\right) = \frac{4}{n^2}\sum DX_i = \frac{4}{n^2} nD\xi =$$

$$= \frac{\theta^2}{3n} \xrightarrow[n\to\infty]{} 0$$

$\forall \theta > 0$ Сост. по дост. усл.

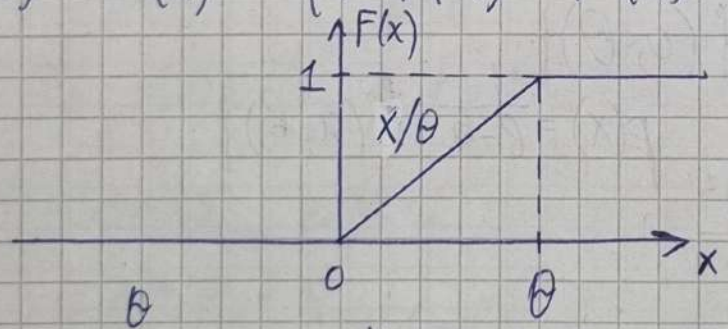
② $\tilde{\theta}_2 = X_{min}$

$\forall \theta > 0 \quad M\tilde{\theta}_2 = MX_{min}$

$\xi \sim F(x)$

$$\xi_{min} \sim \underbrace{1-(1-F(x))^n}_{\Phi}$$

$$\varphi(x) = \Phi'(x) = n(1-F(x))^{n-1}F'(x) = n\left(1-\frac{x}{\theta}\right)^{n-1}\frac{1}{\theta}\{(0,\theta)\}$$



$$MX_{min} = \int_0^\theta xn\underbrace{\left(1-\frac{x}{\theta}\right)}_{t}^{n-1}\frac{1}{\theta}dx = \left\{x = \theta(1-t)\right\} =$$

$$= -\int_1^0 (1-t)n\theta t^{n-1}dt = n\theta\int_0^1 (t^{n-1}-t^n)dt = n\theta\left(\frac{1}{n}-\frac{1}{n+1}\right) = \frac{\theta}{n+1}$$

Смещ.

$\tilde{\theta}_2' = (n+1)X_{min}$

$M\tilde{\theta}_2' = (n+1)MX_{min} = \theta$ Несмещ.

$$D\tilde{\theta}_2' = D((n+1)X_{min}) = (n+1)^2 DX_{min} = \frac{n\theta^2}{n+2} \xrightarrow[n\to\infty]{} 0$$

$$Mx^2_{min} = \int_0^\theta x^2 n \Big(\underbrace{1-\frac{x}{\theta}}_{t}\Big)^{n-1} \frac{1}{\theta}\,dx = \int_0^1 \theta^2 (1-t)^2 n t^{n-1}\,dt =$$

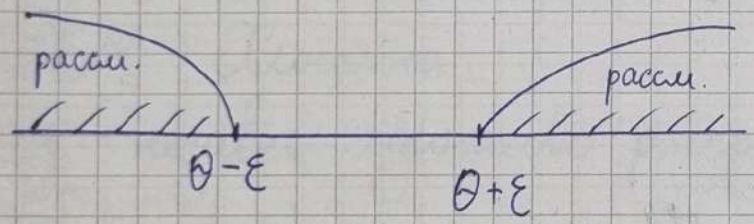
$$= n\theta^2 \int_0^1 \left(t^{n-1} - 2t^n + t^{n+1}\right)dt = n\theta^2\left(\frac{1}{n} - \frac{2}{n+1} + \frac{1}{n+2}\right) = \frac{2\theta^2}{(n+1)(n+2)}$$

$$Dx_{min} = \frac{2\theta^2}{(n+1)(n+2)} - \frac{\theta^2}{(n+1)^2} = \theta^2\left(\frac{2(n+1)-(n+2)}{(n+1)^2(n+2)}\right) =$$

$$= \frac{n\theta^2}{(n+1)^2(n+2)}$$

Достат. усл. не раб.

$$\forall\, \theta>0\ \forall\, \varepsilon>0\quad P\left(|\tilde{\theta}_2' - \theta| \geq \varepsilon\right) \xrightarrow[n\to\infty]{} 0$$



$$P\left(|\tilde{\theta}_2' - \theta| \geq \varepsilon\right) \geq P\left(\tilde{\theta}_1' \geq \theta+\varepsilon\right) =$$

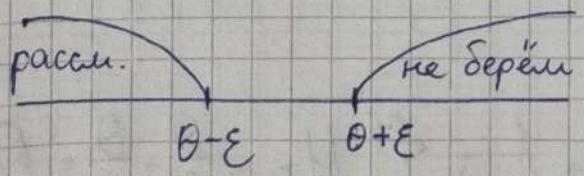
$$= P\left((n+1)x_{min} \geq \theta+\varepsilon\right) = P\left(x_{min} \geq \frac{\theta+\varepsilon}{n+1}\right) =$$

$$= 1 - \Phi\left(\frac{\theta+\varepsilon}{n+1}\right) \underset{n\geq N}{=} 1 - \left(1 - \left(1 - \frac{\theta+\varepsilon}{(n+1)\theta}\right)^n\right) = \left(1 - \frac{\theta+\varepsilon}{(n+1)\theta}\right)^n \xrightarrow[n\to\infty]{} e^{-\frac{n+\varepsilon}{\theta}} > 0$$

$$\Phi(x) = 1 - (1 - F(x))^n$$

Не явл. состоятельной.

$$P\left(|x_{min} - \theta| \geq \varepsilon\right) = P\left(x_{min} \leq \theta - \varepsilon\right) = \Phi(\theta - \varepsilon) =$$



$$P\left(x_{min} \geq \theta + \varepsilon\right) = 0$$

$$= 1 - \left(1 - F(\theta - \varepsilon)\right)^n = 1 - \left(1 - \frac{\theta-\varepsilon}{\theta}\right)^n = 1 - \left(\frac{\varepsilon}{\theta}\right)^n \xrightarrow[n\to\infty]{} 1$$

$\exists\, \varepsilon > 0 \qquad \frac{\varepsilon}{\theta} < 1$

$\exists\, \theta > 0$

Не явл. состоятельной.

③ $\tilde{\theta}_3 = X_{max}$

$M\tilde{\theta}_3 = MX_{max}$

$$X_{max} \sim \underbrace{(F(x))^n}_{\Psi(x)}$$

$$\Psi(x) = \Psi'(x) = n\left(\frac{x}{\theta}\right)^{n-1}\frac{1}{\theta}\{(0,\theta)\}$$

$$MX_{max} = \int_0^\theta x n\left(\frac{x}{\theta}\right)^{n-1}\frac{1}{\theta}\,dx = \frac{n}{\theta^n}\,\frac{\theta^{n+1}}{n+1} = \frac{n}{n+1}\theta$$

$$\tilde{\theta}_3' = \frac{n+1}{n}X_{max}$$

$$M\tilde{\theta}_3' = \frac{n+1}{n}MX_{max} = \theta$$

Несмещ.

$$D\tilde{\theta}_3' = \left(\frac{n+1}{n}\right)^2 DX_{max}$$

$$MX_{max}^2 = \int_0^\theta x^2 n\left(\frac{x}{\theta}\right)^{n-1}\frac{1}{\theta}\,dx = \frac{n}{\theta^n}\,\frac{\theta^{n+2}}{n+2} = \frac{\theta^2 n}{n+2}$$

$$DX_{max} = \theta^2\left(\frac{n}{n+2} - \frac{n^2}{(n+1)^2}\right) = \theta^2\left(\frac{n(n+1)^2 - n^2(n+2)}{(n+2)(n+1)^2}\right) = \frac{n\theta^2}{(n+2)(n+1)^2}$$

$$D\tilde{\theta}_3' = \left(\frac{n+1}{n}\right)^2 DX_{max} = \frac{\theta^2}{n(n+2)} \xrightarrow[n\to\infty]{} 0.$$

Состоят.

④ $\tilde{\theta}_4 = X_1 + \frac{1}{(n-1)}\sum_{i=2}^{n} X_i$

$$M\tilde{\theta}_4 = MX_1 + \frac{1}{(n-1)}\sum_{i=2}^{n} MX_i = M\xi + \frac{1}{(n-1)}\sum_{i=2}^{n} M\xi = \frac{\theta}{2} + \frac{\theta}{2} = \theta$$

Несмещ.

$$D\tilde{\theta}_4 = DX_1 + \frac{1}{(n-1)^2}\sum_{i=2}^{n} DX_i = \frac{\theta^2}{12} + \frac{1}{(n-1)}\frac{\theta^2}{12} \not\to 0$$

Не вып. достат. усл.

$\tilde{\theta}_4 \xrightarrow{P} \theta$ Не явл. сост.

$$X_1 + \underbrace{\frac{1}{(n-1)}\sum_{i=2}^{n} X_i}_{\xrightarrow{P} M\xi = \frac{\theta}{2}} \xrightarrow{P} \xi + \frac{\theta}{2}$$

$\xi_n \xrightarrow{P} \xi$

$\eta_n \xrightarrow{P} \eta$

$\longrightarrow \xi_n + \eta_n \xrightarrow{P} \xi + \eta$

$\xi_n \eta_n \xrightarrow{P} \xi\eta$

$X_1 \xrightarrow{P} \xi$

ЗБЧ Хинчина:

ξ_n независ. одинаково распр., $\exists\, M\xi_n < +\infty$

$$\longrightarrow \frac{1}{n}\sum_{i=1}^{n} \xi_i \xrightarrow{P} M\xi_1$$

Итого:

$\tilde{\theta}_3' = \frac{n+1}{n} X_{max}$

$\tilde{\theta}_1 = 2\bar{X}$

$D\tilde{\theta}_1 = \frac{\theta^2}{3n}$

$D\tilde{\theta}_3 = \frac{\theta^2}{(n+2)n}$

$\frac{1}{3n}$ $\frac{1}{n(n+2)}$

$n^2 + 2n > 3n$

$n > 1$

$\tilde{\theta}_3'$ более эфф. чем $\tilde{\theta}_1$

По определению:

$$\forall \varepsilon > 0 \quad P(|\tilde{\theta}_3' - \theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0$$

$$P(|\theta_3' - \theta| \geq \varepsilon) = P(\tilde{\theta}_3' - \theta \geq \varepsilon) +$$

$$+ P(\tilde{\theta}_3' \leq \theta - \varepsilon) = P\left(\frac{n+1}{n} X_{max} \leq \theta - \varepsilon\right) =$$

$$= P\left(X_{max} \leq \frac{(\theta-\varepsilon)n}{n+1}\right) = P\left(X_{max} \leq \frac{n(\theta-\varepsilon)}{n+1}\right) =$$

$$= \left(F\left(\frac{n(\theta-\varepsilon)}{n+1}\right)\right)^n = \left(\frac{n(\theta-\varepsilon)}{\theta(n+1)}\right)^n =$$

$$= \left(\left(1-\frac{\varepsilon}{\theta}\right)\frac{n}{n+1}\right)^n \xrightarrow[n\to\infty]{} 0$$

При $\varepsilon > \theta$ $\quad P\left(X_{max} \leq \frac{(\theta-\varepsilon)n}{n+1}\right) = 0$

$$\Rightarrow P\left(X_{max} \leq \frac{n(\theta-\varepsilon)}{n+1}\right) \xrightarrow[n\to\infty]{} 0 \Rightarrow$$

$\Rightarrow \tilde{\theta}_3'$ - состоятельна.

Рассмотрим $\tilde{\theta}_3$, исследуем на состоятельность.

$$\forall \varepsilon > 0 \quad P(|\tilde{\theta}_3 - \theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0$$

$$P(|\tilde{\theta}_3 - \theta| \geq \varepsilon) = P(\tilde{\theta}_3 \geq \varepsilon + \theta) +$$

$$+ P(\tilde{\theta}_3 \leq \theta - \varepsilon) = (F(\theta-\varepsilon))^n = \left(\frac{\theta-\varepsilon}{\theta}\right)^n \xrightarrow[n\to\infty]{\varepsilon<\theta} 0$$

Для $\varepsilon > \theta$:

$$\theta - \varepsilon < 0 \Rightarrow$$

$$\Rightarrow P(\tilde{\theta}_3 \leq \theta - \varepsilon) = 0$$

$\Rightarrow \tilde{\theta}_3$ тоже состоятельна.